Preprocessing — UCI Heart Disease (Cleveland)

- Initial shape: (303 × 14).
- After removing missing values: (293 × 14).

Class distribution of target column

B4 making it a binary column

```
Counts:
 target
0    160
1     54
2     35
3     35
4     13
Name: count, dtype: int64

Percentage:
 target
0    53.872054
1    18.181818
2    11.784512
3    11.784512
4     4.377104
Name: proportion, dtype: float64
```

After making it a binary column

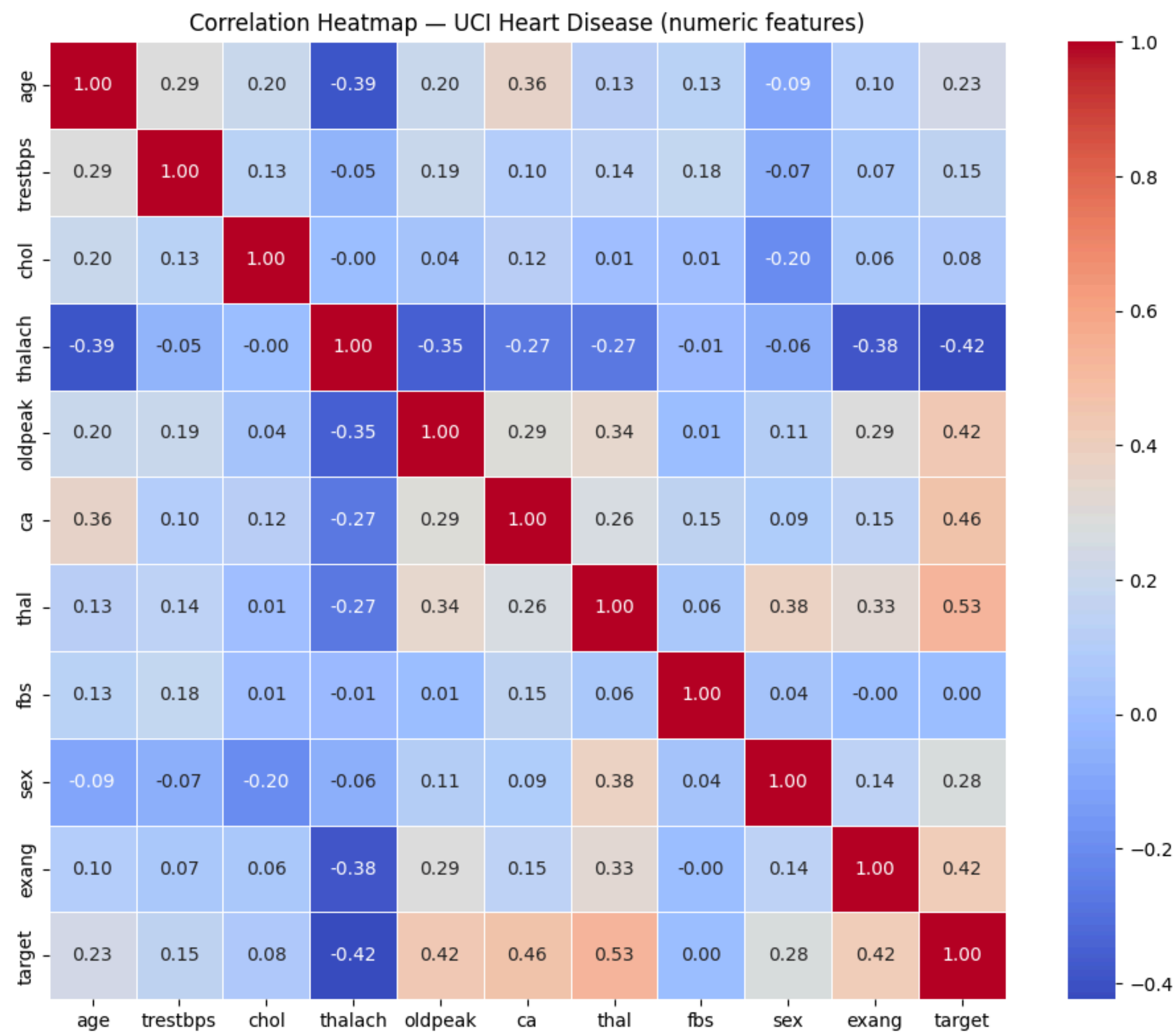
```
Counts:
 target
0    160
1    137
Name: count, dtype: int64

Percentage:
 target
0    53.872054
1    46.127946
Name: proportion, dtype: float64
```

Since the imbalance is minor, applying Synthetic Minority Over-sampling Technique (SMOTE) is not necessary, as it could artificially introduce synthetic samples and potentially distort the natural distribution of clinical data.

Instead, class weighting is more appropriate because:

- It preserves the original dataset distribution
- It avoids introducing synthetic data
- It is computationally efficient
- It is well-suited for small to moderate imbalance



Top 3 strongest feature pair correlations:

thal target 0.526640

ca target 0.463189

oldpeak target 0.424052

dtype: float64

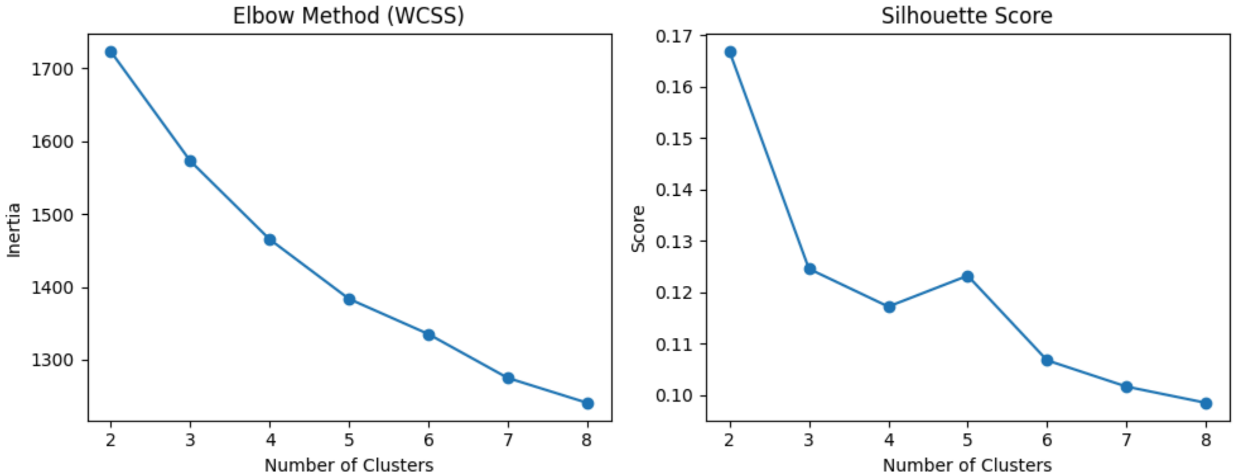
These correlations can cause issues for a Naive Bayes classifier because Naive Bayes assumes that all features are independent of each other, but here we clearly see that some features are related. When this assumption is violated, the model may give biased probability estimates and slightly weaker performance, especially for medically related variables like exercise-induced changes and vessel count.

- Added OHE on categorical columns
- Added standard scaler on numerical/continuous columns
- Stratifies 80/20 train test split

Part A: Unsupervised Learning

K Means

- K-Means for k ∈ {2, 3, 4, 5, 6, 7, 8} on the full standardised feature matrix



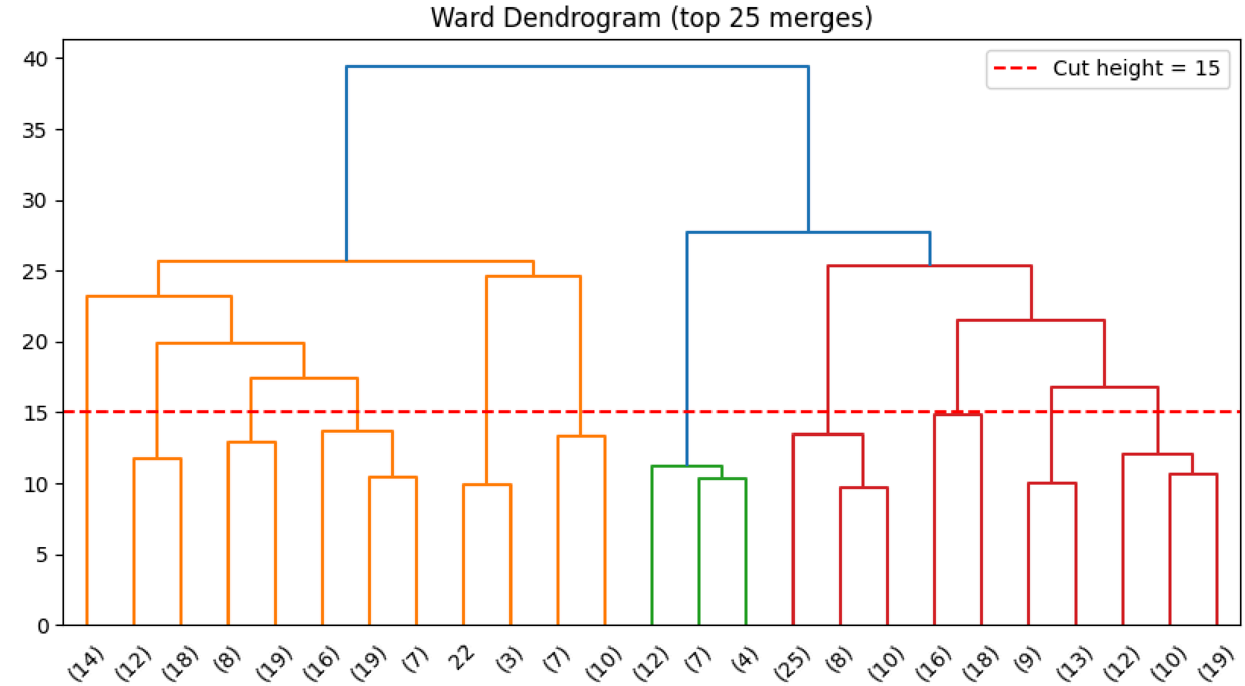
- We choose k based on:
  - elbow bend in WCSS
  - highest silhouette score
- **Chosen k = 2.**
- the silhouette score is highest at k = 2
- The K-Means clusters show only partial alignment with the clinical ground truth, as some clusters are dominated by a single class, but there is still noticeable mixing of diseased and non-diseased patients within clusters.

```
Cluster 0 | Size: 152 | Disease %: 48.7%
thalach    151.072368
oldpeak      1.053947
cp           3.210526
dtype: float64

Cluster 1 | Size: 145 | Disease %: 43.4%
thalach    148.055172
oldpeak      1.057241
cp           3.103448
```

- ARI: 0.3776
- An ARI score of 0.3776 means the clusters and actual labels match a little, but not very well. This shows that the data has some grouping, but it is not clear. People with and without heart disease are mixed together, so clustering cannot separate them perfectly.

Hierarchical Clustering

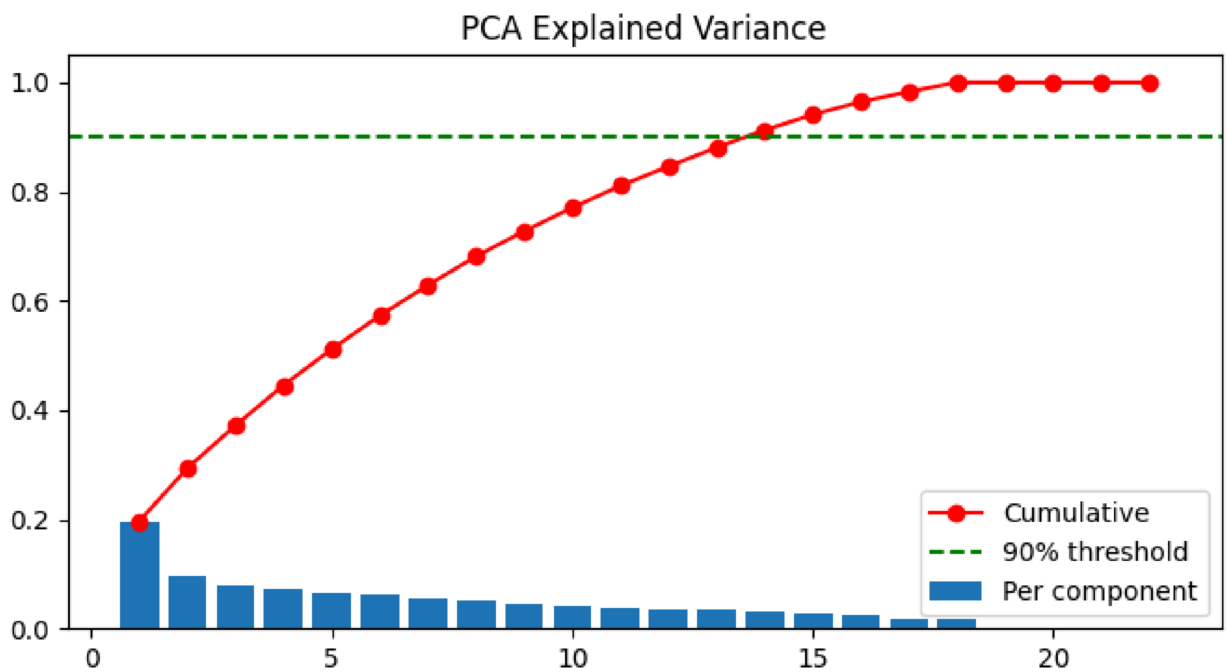


	No Disease (0)	Disease Present (1)
Cluster		
0	127	36
1	33	101

36 disease positive people occur in cluster if disease free people whereas 33 disease negative people are present in disease positive cluster

- An ARI of 0.4562 means the two methods agree somewhat, but not fully. They find similar groups, but there are still differences. I would trust hierarchical clustering more because it is more stable and does not depend on random starting points like K-Means.

Dimensionality Reduction



Components needed for 90% variance: 14

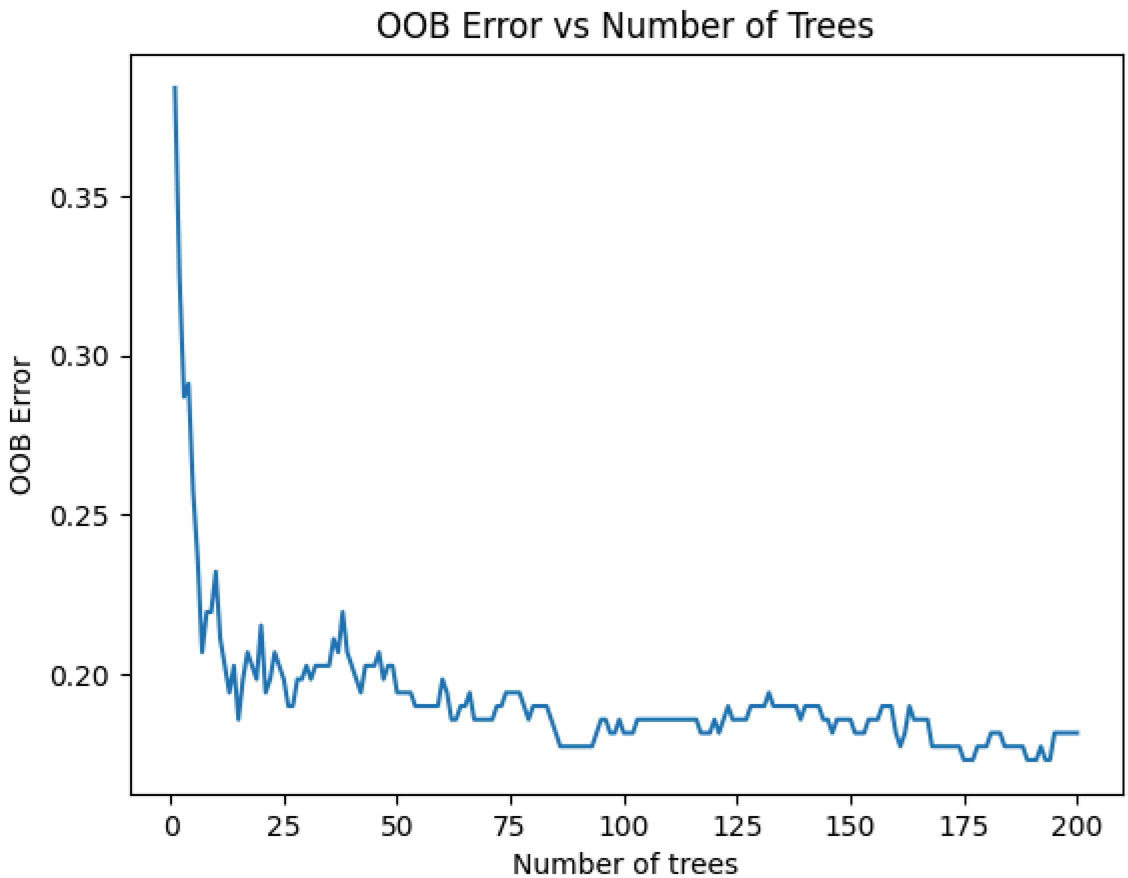
- The two components appear somewhat separable in the visualisation, bit some are mixed together showing the class of classifying over here is not as easy.

Part B: Bagging and Boosting

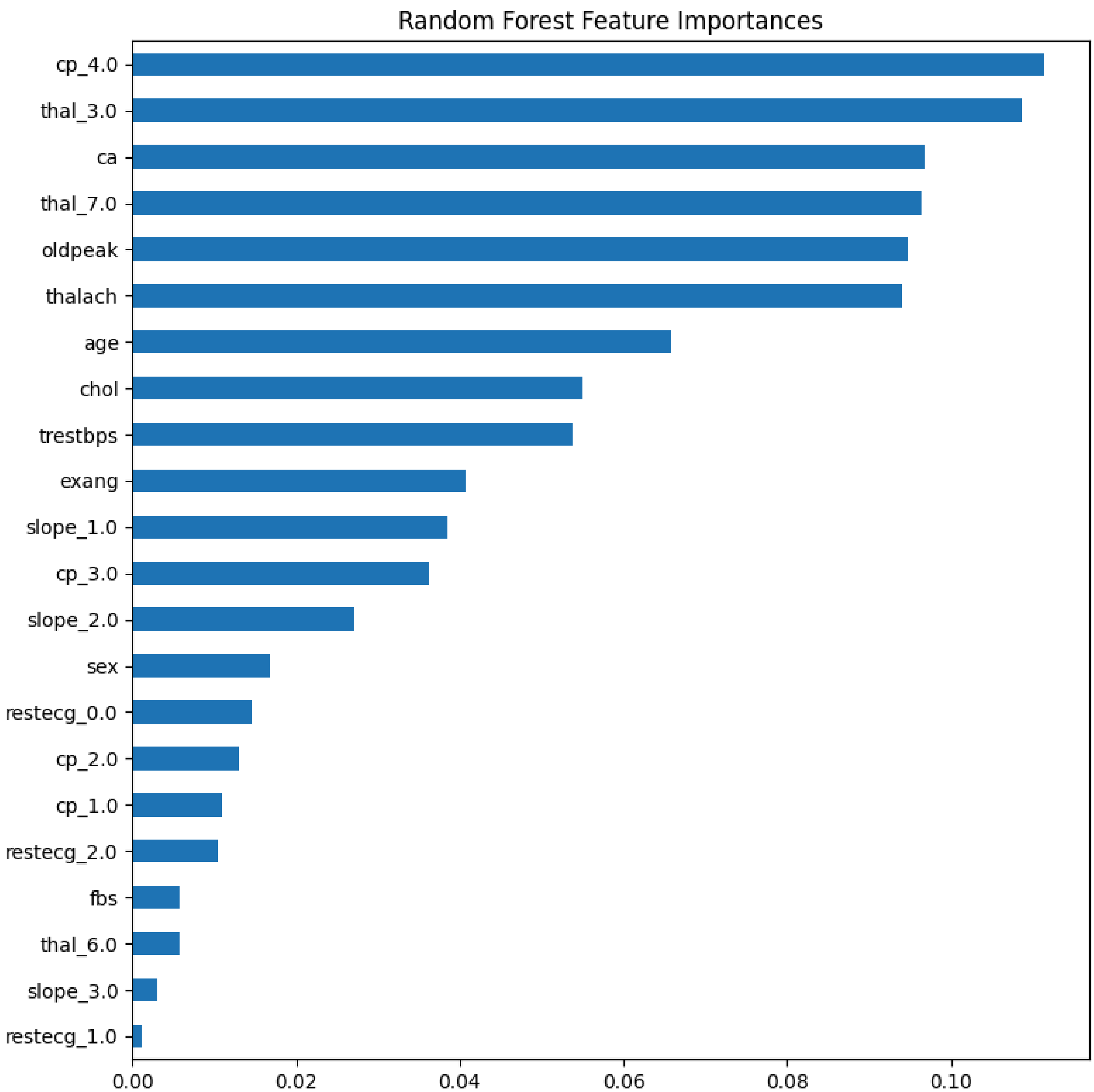
Random Forest

- Best params: {'max_depth': 5, 'n_estimators': 200}

  Best CV F1: 0.7895



Yes this matches the best chosen n_Estimator and max_depth combination adn OOB Error starts becoming flat there.



Top 5 features:

cp_4.0	0.111327
thal_3.0	0.108700
ca	0.096759
thal_7.0	0.096380
oldpeak	0.094654

cp_4.0: This indicates asymptomatic chest pain, which is strongly associated with silent or advanced coronary artery disease.

thal_3.0: Represents normal thalassemia results, and deviations in thal values are linked to abnormal blood flow and heart dysfunction.

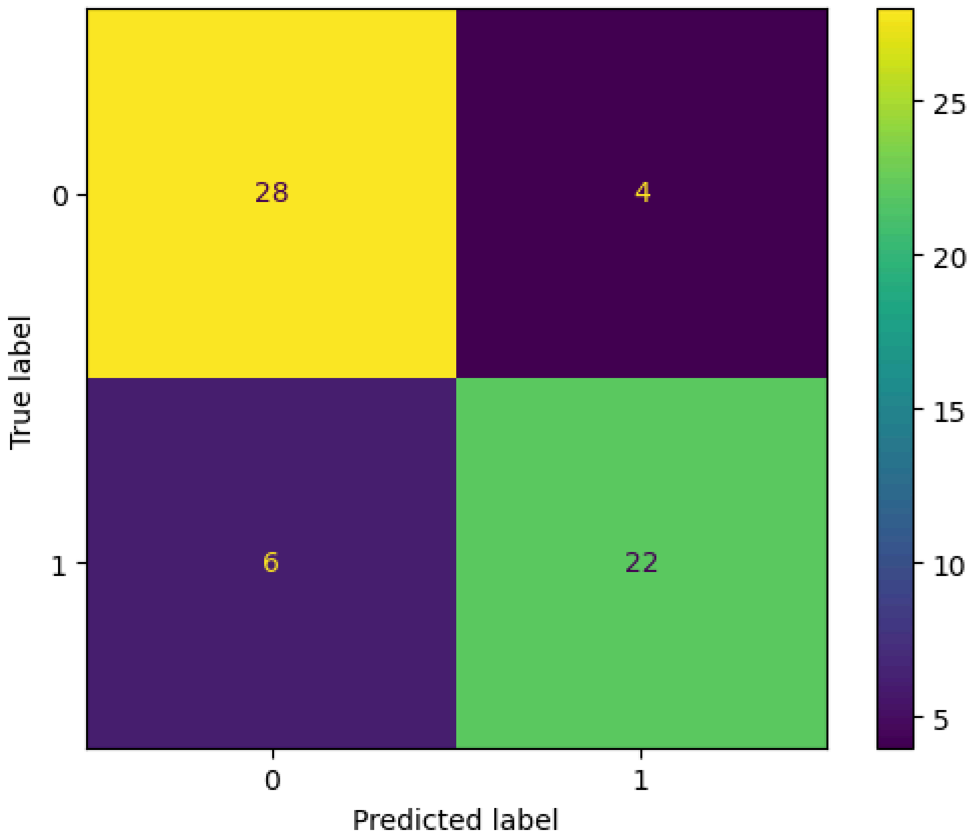
ca: Number of major vessels colored by fluoroscopy, where higher values indicate blocked arteries and increased heart disease risk.

thal_7.0: Represents reversible defect in thalassemia scan, indicating temporary blood flow issues due to coronary artery narrowing.

oldpeak: Measures ST depression during exercise, where higher values indicate greater cardiac stress and likelihood of ischemia.

	precision	recall	f1-score	support
0	0.82	0.88	0.85	32
1	0.85	0.79	0.81	28
accuracy			0.83	60
macro avg	0.83	0.83	0.83	60
weighted avg	0.83	0.83	0.83	60

AUC-ROC: 0.933

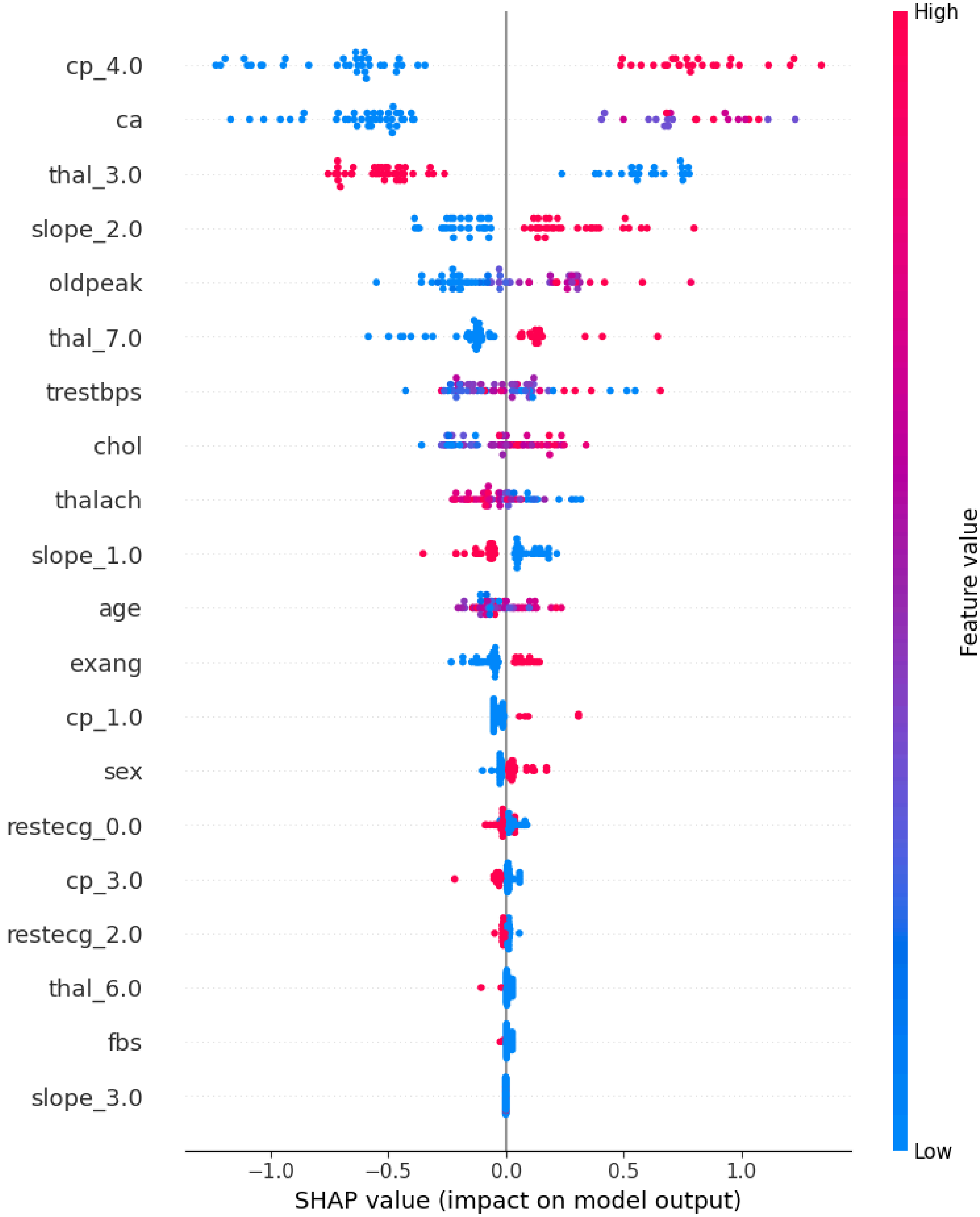


Class 1 (disease present) has worse recall (0.79) compared to class 0 (0.88), meaning the model is missing more actual heart disease cases.

In a cardiac screening context, the false negatives (6 cases) are more critical because they represent patients who actually have heart disease but are incorrectly predicted as healthy, which can delay diagnosis and treatment and increase the risk of severe cardiac events.

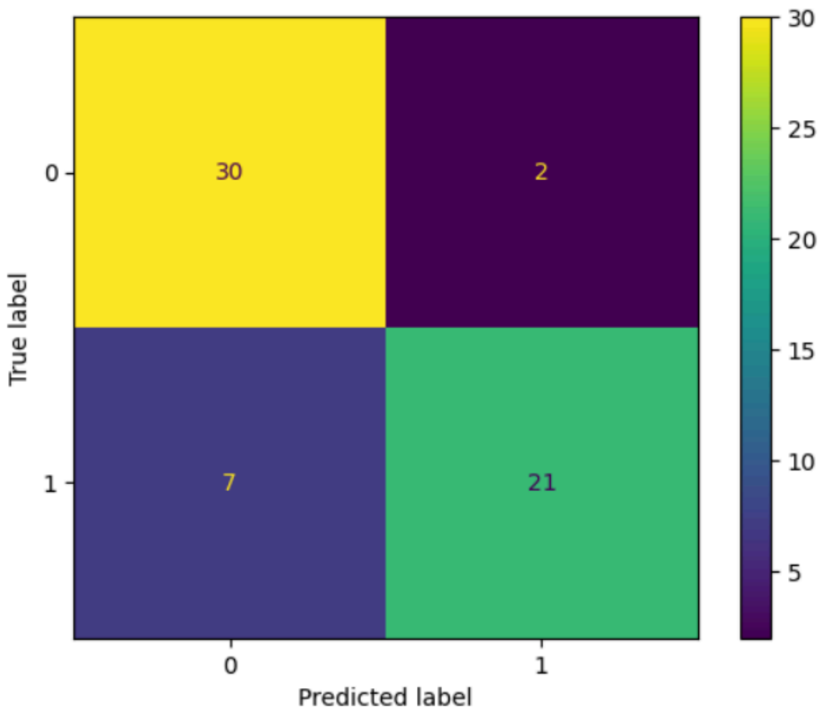
XGBoost

- Best params: {'learning_rate': 0.3, 'max_depth': 5} Best CV F1: 0.7897
- Yes overfitting is there as model performed too well on testing data but on validation set it fails as the loss doesn't decrease as much.



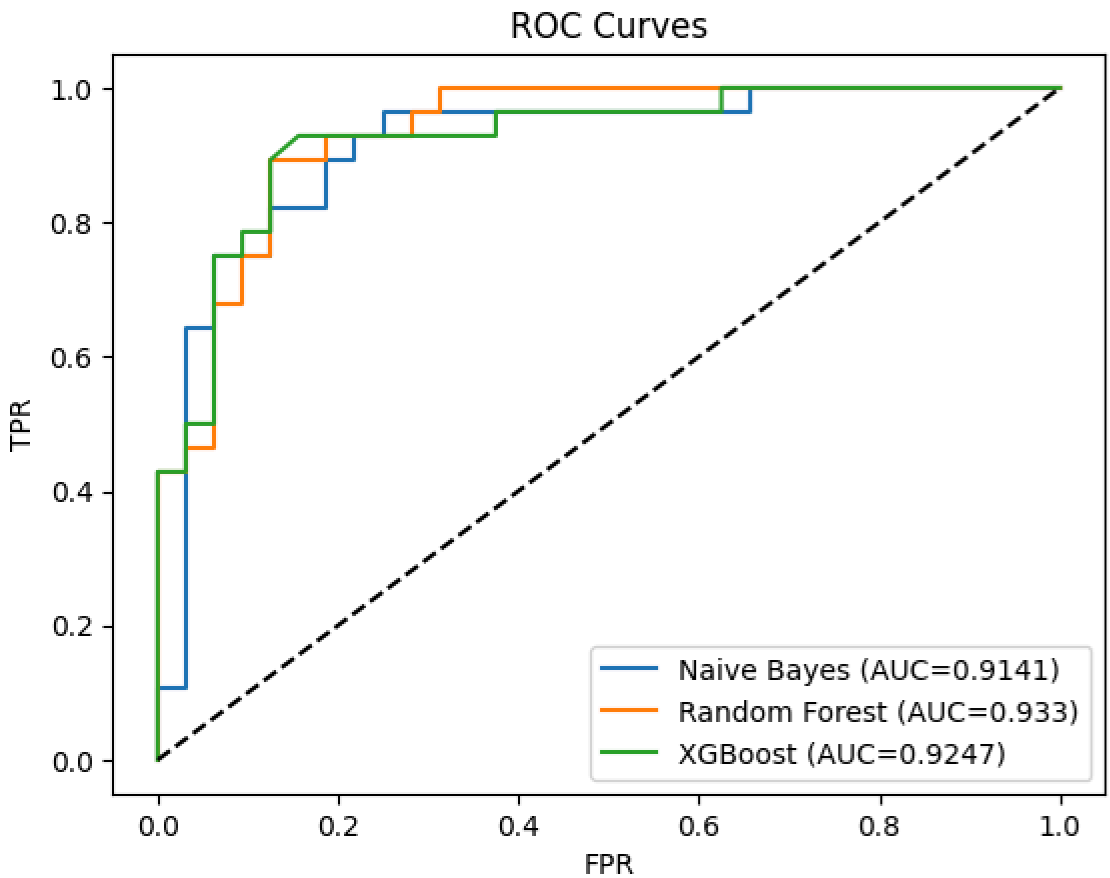
cp_4.0
ca
thal_3.0
slope_2.0
oldpeak
thal_7.0
trestbps
chol
thalach
slope_1.0
age
exang
cp_1.0
sex
restecg_0.0
cp_3.0
restecg_2.0
thal_6.0
fbs
slope_3.0
−1.0
−0.5
0.0
0.5
1.0
SHAP value (impact on model output)
High
Low
Feature value

	precision	recall	f1-score	support
0	0.81	0.94	0.87	32
1	0.91	0.75	0.82	28
accuracy			0.85	60
macro avg	0.86	0.84	0.85	60
weighted avg	0.86	0.85	0.85	60



Ensemble Comparison & ROC

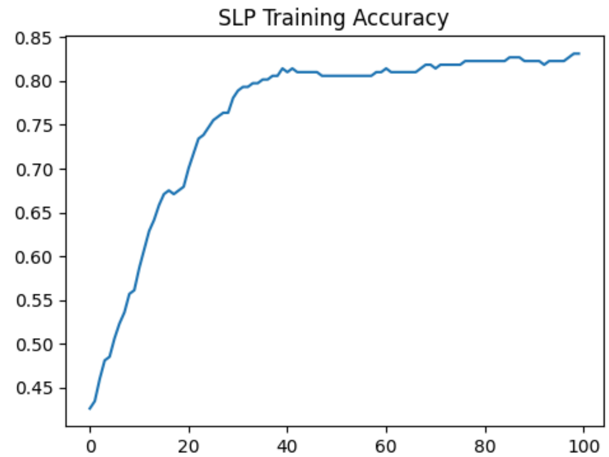
	Classifier	Accuracy	Macro F1	AUC-ROC	Recall (Disease)	Train Time (s)
0	Random Forest	0.8333	0.8316	0.9330	0.7857	0.0642
1	XGBoost	0.8500	0.8465	0.9247	0.7500	0.0102
2	Naive Bayes	0.8500	0.8490	0.9141	0.8214	0.0048



Naive Bayes would be the most suitable model for deployment in a community hospital screening pipeline. Although Random Forest has the highest AUC-ROC (0.933), Naive Bayes achieves the best recall for the disease class (0.8214), meaning it correctly identifies more patients with heart disease. In medical screening, recall is more important than overall accuracy because missing a positive case (false negative) can delay treatment and lead to severe or life-threatening outcomes. Naive Bayes is also highly interpretable and computationally lightweight, making it practical for low-resource clinical settings. Therefore, despite slightly lower AUC, its higher sensitivity and simplicity make it the safest choice for real-world screening.

Part C: Artificial Neural Networks on Tabular Data

Single-Layer Perceptron (SLP)



Top 3 features by SLP weight:

cp_4.0: 0.6904

thal_7.0: 0.6702

thal_3.0: -0.6483

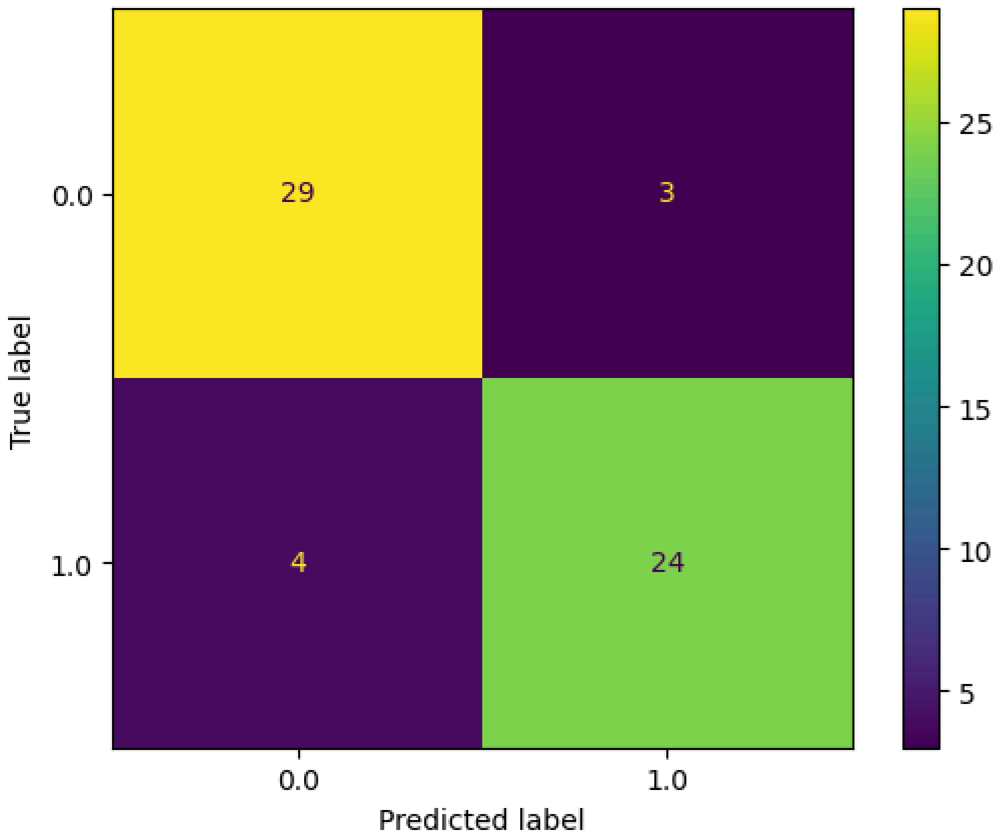
-> Thal 7.0 comes at 4th place in Random forest whereas over here it comes in 2nd place, but other 2 positions match so yes we can say they almost match.

```
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 76ms/step
              precision    recall  f1-score   support

         0.0       0.88      0.91      0.89        32
         1.0       0.89      0.86      0.87        28

    accuracy                           0.88        60
   macro avg       0.88      0.88      0.88        60
weighted avg       0.88      0.88      0.88        60

AUC: 0.9241
```



A linear model is limited because it can only draw straight-line decision boundaries. Heart disease prediction depends on complex, non-linear relationships between features. Because of this, a single-layer perceptron cannot fully capture the patterns in the data.

Multi-Layer Perceptron (MLP)

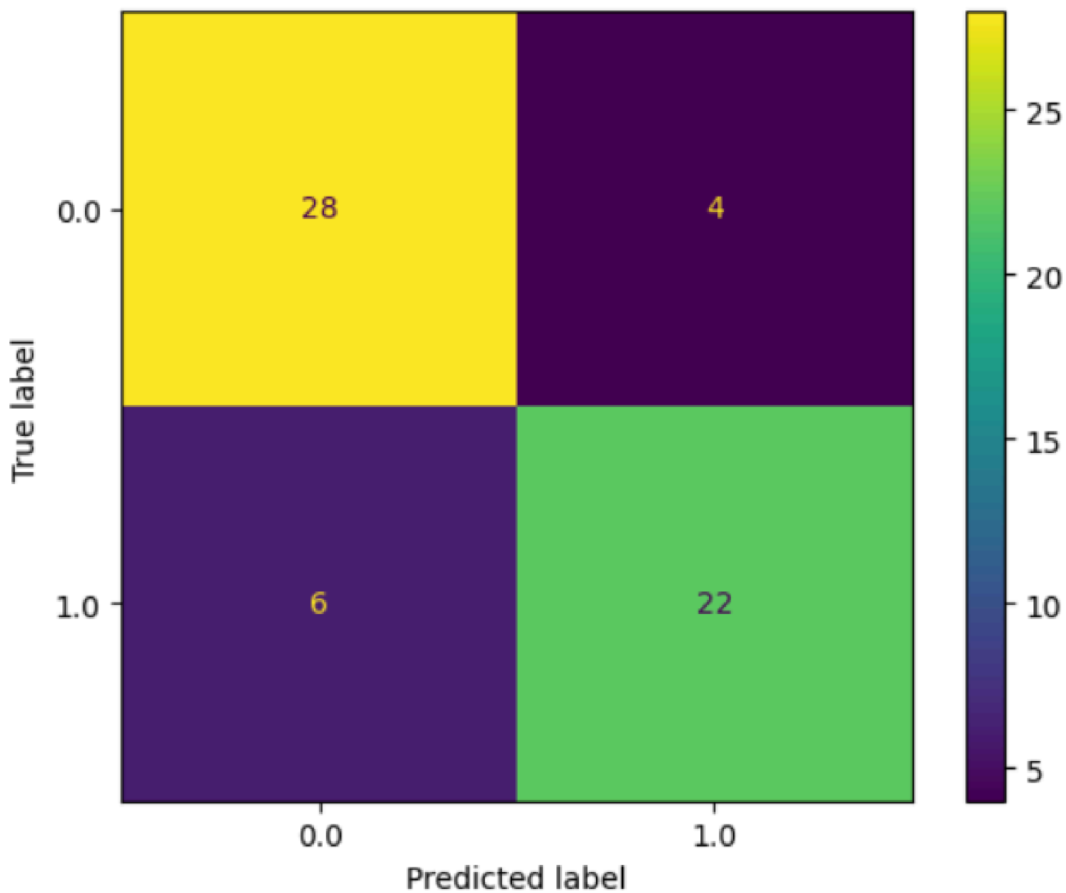
- Activation function: ReLU was used in all hidden layers because it is computationally efficient and helps reduce vanishing gradient issues, making training stable for tabular data.
- Regularisation strategy: Dropout (0.3) was applied after each hidden layer to reduce overfitting by randomly deactivating neurons during training.
- Optimizer and learning rate: Adam optimizer was used with its default learning rate (0.001) because it adapts learning rates automatically and works well for small datasets.
- Number of epochs: The model was trained for up to 150 epochs, but EarlyStopping with patience=10 on validation loss was used to stop training early once performance stopped improving.

```
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 56ms/step
              precision    recall  f1-score   support

         0.0       0.82      0.88      0.85        32
         1.0       0.85      0.79      0.81        28

    accuracy                           0.83        60
   macro avg       0.83      0.83      0.83        60
weighted avg       0.83      0.83      0.83        60

AUC: 0.9442
```



The MLP achieves strong performance with an AUC-ROC of 0.9442, which is higher than all ensemble models, indicating better overall ranking ability for distinguishing between classes. However, its accuracy (0.83) and recall for the disease class (0.79) are slightly lower than Naive Bayes and comparable to Random Forest. Among ensemble methods, Naive Bayes shows the highest recall (0.8214), making it better at catching disease cases. Overall, while the MLP is powerful in terms of AUC, the ensembles especially Naive Bayes are more reliable for minimizing false negatives in a clinical setting.

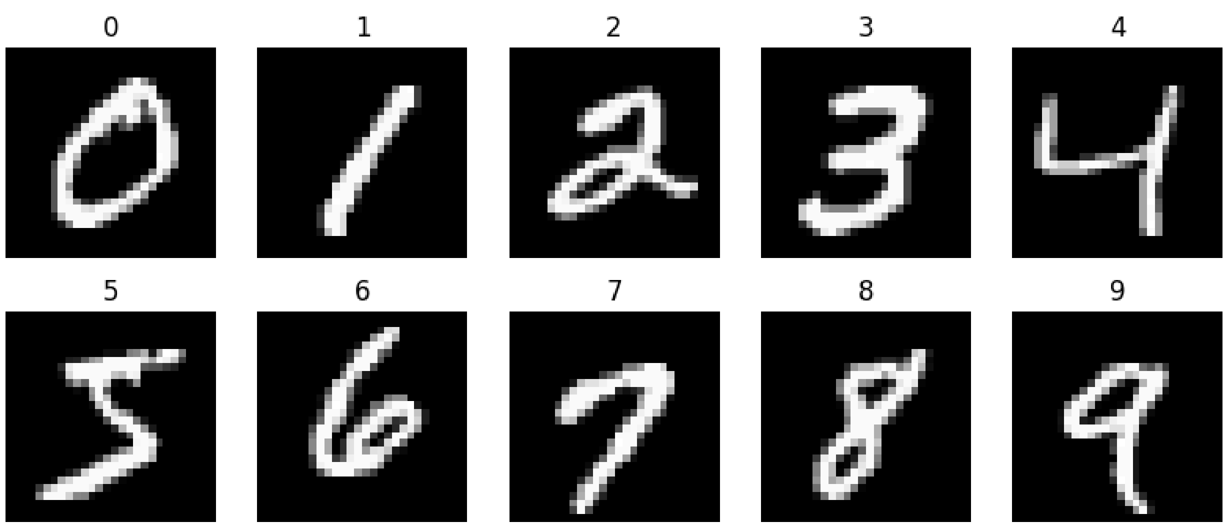
Ablation Study

The activation function is the most important component. Replacing ReLU with Sigmoid causes a large drop in performance (0.4516), showing it cannot learn the patterns well. Dropout and EarlyStopping only slightly affect results and mainly improve generalisation. Overall, activation choice has the biggest impact on MLP performance.

	Test F1
Best MLP	0.8148
A: No Dropout	0.8077
B: Sigmoid	0.4516
C: No EarlyStopping	0.8214

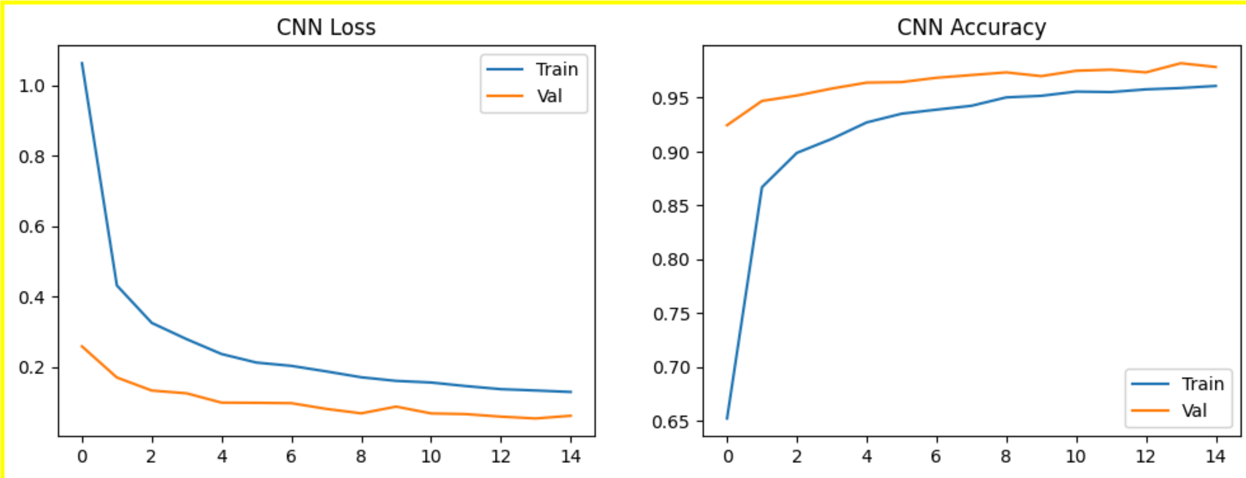
Part D: CNN on MNIST Digit Images

Data Preparation & Baseline



MLP Baseline Test Accuracy: 0.9065

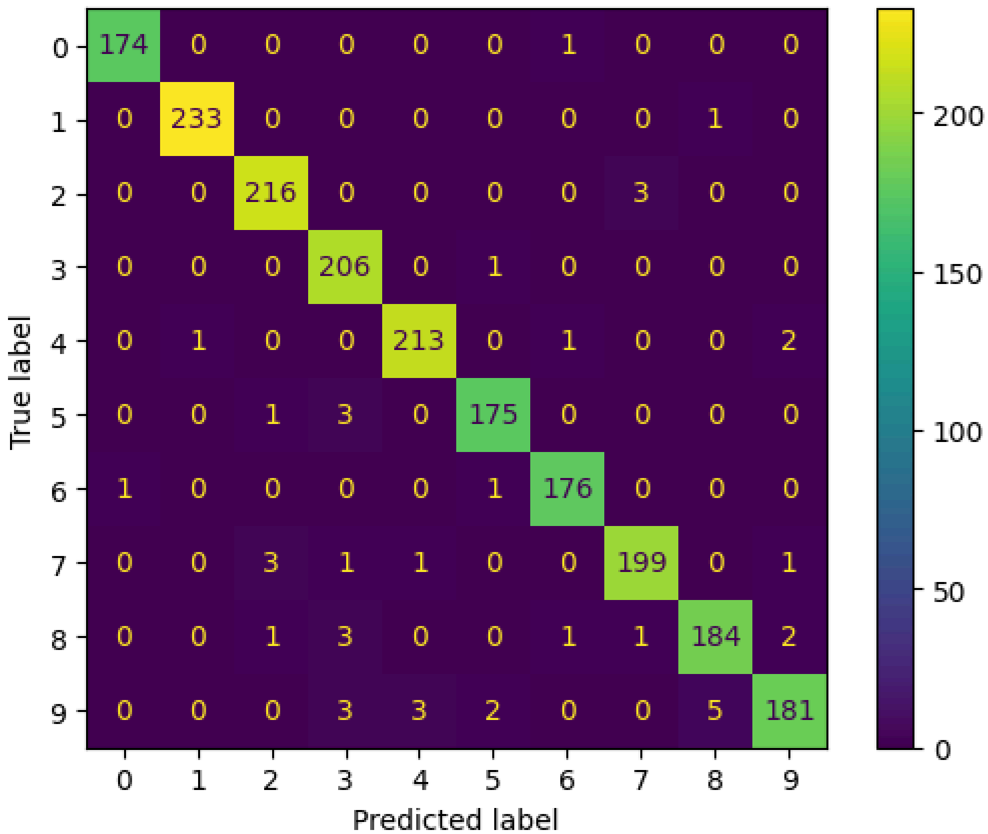
Lightweight CNN




```
••• 63/63 ━━━━━━━━━━━━━━━━━━━━ 1s 14ms/step
              precision    recall  f1-score   support

           0       0.99      0.99      0.99       175
           1       1.00      1.00      1.00       234
           2       0.98      0.99      0.98       219
           3       0.95      1.00      0.97       207
           4       0.98      0.98      0.98       217
           5       0.98      0.98      0.98       179
           6       0.98      0.99      0.99       178
           7       0.98      0.97      0.98       205
           8       0.97      0.96      0.96       192
           9       0.97      0.93      0.95       194

    accuracy                           0.98      2000
   macro avg       0.98      0.98      0.98      2000
weighted avg       0.98      0.98      0.98      2000
```



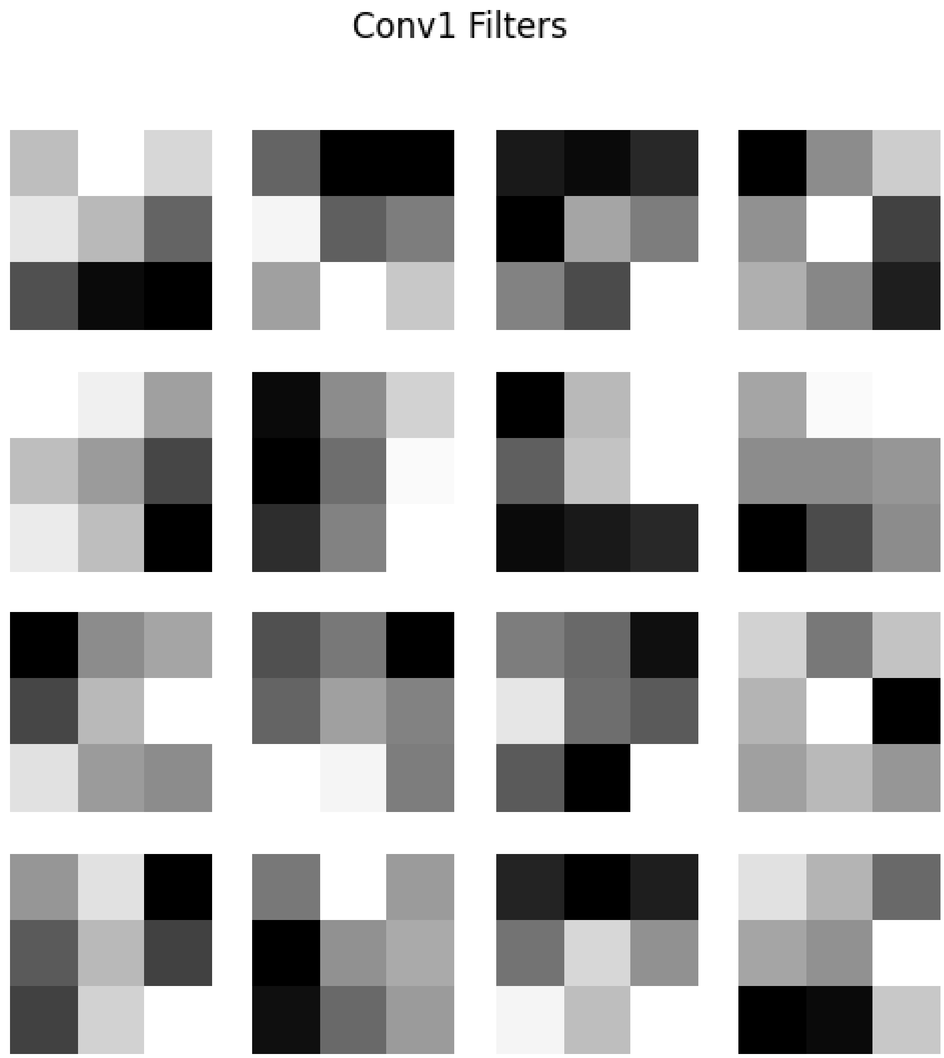
Most confused digit pair: (np.int64(9), np.int64(8))

Number of misclassifications: 5

Most Confused Digit Pairs
The most confused digit pair is 9 and 8 with 5 misclassifications. These digits are difficult to distinguish because they share similar circular structures and overlapping loops. In many handwritten samples, slight variations in writing style make 8 resemble a closed 9 or vice versa, leading to confusion in feature extraction by the CNN.

CNN vs MLP Comparison (Epochs)
The MLP baseline achieves a test accuracy of 0.9065 after 5 epochs. The CNN reaches a much higher accuracy of 0.98 and surpasses the MLP's final accuracy within the first epoch, where validation accuracy already exceeds 0.92. This shows that CNNs learn spatial features from images much more efficiently than fully connected MLPs.

Visualising What the CNN Learned



The 16 filters in the first Conv2D layer learn simple patterns like edges, lines, and curves. Some detect vertical lines, some horizontal lines, and some diagonal strokes. They help the model recognize parts of digits, not brightness. These small patterns are later combined to understand full digits.

Feature Map Observation
Each feature map responds to simple visual patterns in the digit image such as edges, curves, and strokes. Some channels detect vertical or horizontal lines, while others respond to curved parts of digits, helping the model break the image into basic shapes.

Model Trust and CNN vs Fully Connected Networks
These visualisations help build trust because they show what the model is focusing on when making predictions instead of acting like a black box. We can see that the CNN is learning meaningful patterns like edges and shapes rather than random pixel values. This makes it easier to understand why certain digits are classified correctly or confused. Unlike fully

connected networks, CNNs preserve spatial structure, allowing them to learn local features from images more effectively.